

3*-Connectivity of the Fully Connected Cubic Networks

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Abstract

A k -container $C(u, v)$ of G between u and v is a set of k internally disjoint paths between u and v . A k -container $C(u, v)$ of G is a k^* -container if it contains all vertices of G . A graph G is k^* -connected if there exists a k^* -container between any two distinct vertices. In this paper, we prove that the n -dimensional fully connected cubic network $FCCN_n$ is 3*-connected for $n \geq 2$.

1 Introduction

As is customary in structure studies of parallel architectures, we focus on a set of identical processors and view the architectures of the underlying interconnection networks as graphs. The vertices of a graph represent the processors of architecture, and the edges of the graph represent the communication links between processors. Network topology is a crucial factor for the interconnection networks since it determines the performance of the networks. Many interconnection network topologies have been proposed in the literature for the purpose of connecting thousands of processing elements. The hypercube networks have received much attention over the past few years since they offer a rich interconnection structure with large bandwidth, logarithmic diameter, high degree of fault tolerance, and embedding of various interconnection. The hypercube network has been chosen as the interconnection topology in several commercially available parallel computers [11]. However, the hypercube networks are not truly expandable because we have to change the hardware configuration of all the vertices whenever the number of vertices grows exponentially, as the vertices have to be provided with additional ports. Fortunately, hierarchical interconnection networks (HINs) provide excellent expandability to meet this need.

For decades, two kinds of HINs were proposed

addressing the issue of constructing hierarchical interconnection networks. One is the HINs consisting of exactly two levels [3, 4, 8, 12, 13, 15], and another is recursively defined HIN [2, 16, 18, 19]. Moreover, the fully connected cubic networks (FCCNs) proposed in [16] are defined recursively by taking the 3-dimensional cube as the basis graph. It indicates that fully connected cube networks are a class of newly proposed hierarchical networks for multisystems, which enjoy the strengths of constant vertex degree and good expandability. Some interesting properties about FCCNs are discussed in [16]. In particular, Yang et al. [17] presented the shortest-path routing algorithm, Ho et al. [5] find the values of fault-tolerant hamiltonian connectivity and fault-tolerant hamiltonicity. Super spanning connectivity is an important issues in the design and the analysis of interconnection networks [7, 9, 14]. In this paper, we discuss this parameter for FCCNs.

The remaining of this paper is organized as follows: Section 2 provides the formal definition of FCCNs and some preliminaries. In Section 3, we give our main result.

2 Preliminaries

An interconnection network is represented by a graph $G = (V, E)$ with vertices and edges symbolizing the processors and communication links between processors, respectively. For the graph definitions and notations we follow [1]. $G = (V, E)$ is a *graph* if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. A graph H is a *subgraph* of graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Two vertices u and v are *adjacent* if (u, v) is an edge of G . The set of *neighbors* of u , $N(u)$, is $\{v \mid (u, v) \in E\}$. The *degree* of a vertex u of G , $\deg_G(u)$, is the number of edges incident with u . We use $\delta(G)$ to denote the minimum degree in G . A graph G is k -regular if $\deg_G(u) = k$

for every vertex u in G . A *path* is a sequence of vertices represented by $\langle v_0, v_1, \dots, v_k \rangle$ with no repeated vertices and (v_i, v_{i+1}) is an edge of G for all $0 \leq i \leq k-1$ when $k \geq 1$ and $\langle v_0 \rangle$ when $k = 0$. We also write the path $P = \langle v_0, v_1, \dots, v_k \rangle$ as $\langle v_0, \dots, v_i, Q, v_j, \dots, v_k \rangle$, where Q is a path from v_i to v_j . We use P^{-1} to denote the path $\langle v_k, v_{k-1}, \dots, v_0 \rangle$. The *length* of a path Q , $l(Q)$, is the number of edges in Q . The *distance* of vertices u and v of G , $d_G(u, v)$, is the length of the shortest path between u and v . A path is a *hamiltonian path* if it contains all vertices of G . A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two distinct vertices of G . A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every vertex of G . A graph is *hamiltonian* if it has a hamiltonian cycle.

A k -*container* $C(u, v)$ of G between u and v is a set of k *internally disjoint paths* between u and v . The concept of container is proposed by Hsu [6] to evaluate the performance of communication of an interconnection network. The *connectivity* of G , $\kappa(G)$, is the minimum number of vertices whose removal leaves the remaining graph disconnected or trivial. It follows from Menger's Theorem [10] that there is a k -container between any two distinct vertices of G if G is k -connected.

In this paper, we are interested in specified containers. A k -container $C(u, v)$ of G is a k^* -*container* if it contains all vertices of G . A graph G is k^* -*connected* if there exists a k^* -container between any two distinct vertices. An 1^* -connected graph is actually a hamiltonian connected graph. Moreover, a 2^* -connected graph is a hamiltonian graph. Thus, the concept of k^* -connected graph is a hybrid concept of connectivity and hamiltonicity. We define the *spanning connectivity* of a graph G , $\kappa^*(G)$, to be the largest integer k such that G is w^* -connected for all $1 \leq w \leq k$ if G is an 1^* -connected graph. A graph G is *super spanning connected* if $\kappa^*(G) = \kappa(G)$. Obviously, the complete graph K_n is super spanning connected.

From the application point of view, k^* -containers can be used in multipath communication. Note that graph connectivity and hamiltonicity are two very interesting topics in graph theory. The concept of spanning connectivity is interesting as well. From the theoretical point of view, we should put as much emphasis as we did

for hamiltonian graphs. Recently, several families of interconnection networks are proved to be super spanning connected [7, 9, 14]. In this paper, we prove that the n -dimensional fully connected cubic network $FCCN_n$ is super spanning connected.

Let $Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$. For $m \geq 1$ and $a \in Z_8$, let $a^m = \underbrace{aa \dots a}_m$.

For $n \geq 1$, an n -level $FCCN$, $FCCN_n$, is a graph defined recursively as follows:

(1) $FCCN_1$ is a graph with $V(FCCN_1) = Z_8$ and $E(FCCN_1) = \{(0, 1), (0, 2), (1, 3), (2, 3), (4, 5), (4, 6), (5, 7), (6, 7), (0, 4), (1, 5), (2, 6), (3, 7)\}$. Obviously, $FCCN_1$ is isomorphic to Q_3 .

(2) When $n \geq 2$, $FCCN_n$ is built from eight vertex-disjoint copies of $FCCN_{n-1}$ by adding 28 edges. For $0 \leq k \leq 7$, we let $kFCCN_{n-1}$ denote a copy of $FCCN_{n-1}$ with each vertex being prefixed with k , then $FCCN_n$ is defined by

$$V(FCCN_n) = \bigcup_{k=0}^7 V(kFCCN_{n-1}),$$

$$E(FCCN_n) = \left(\bigcup_{k=0}^7 E(kFCCN_{n-1}) \right) \cup$$

$$\{(pq^{n-1}, qp^{n-1}) \mid 0 \leq p < q \leq 7\}.$$

Let $n \geq 2$. A vertex v in $FCCN_n$ is a *boundary vertex* if it is of the form p^n and v is an *intercubic vertex* if it is of the form pq^{n-1} with $p \neq q$. An *intercubic edge* is an edge joining two intercubic vertices.

Figure 1(a), (b), and (c) show $FCCN_1$, $FCCN_2$ and $FCCN_3$. In essence, each vertex of an $FCCN$ has four links, with each boundary vertex having one I/O channel link that is not counted in the vertex degree. Obviously, $kFCCN_{n-1}$ have 7 intercubic vertices and 1 boundary vertex for $0 \leq k \leq 7$ and $n \geq 2$. It is easy to check that if we shrink each $FCCN_{n-1}$ in $FCCN_n$ to a vertex, we will get a complete graph K_8 . We call it be a complete property.

In order to discuss the super spanning connectivity of fully connected cubic networks, we need to introduce the extended fully connected cubic networks. For every $t \in Z_8$, the *extended fully connected cubic network* $FCCN_n^{r,s,t}$ is the graph obtained from $FCCN_n$ by joining the vertices in

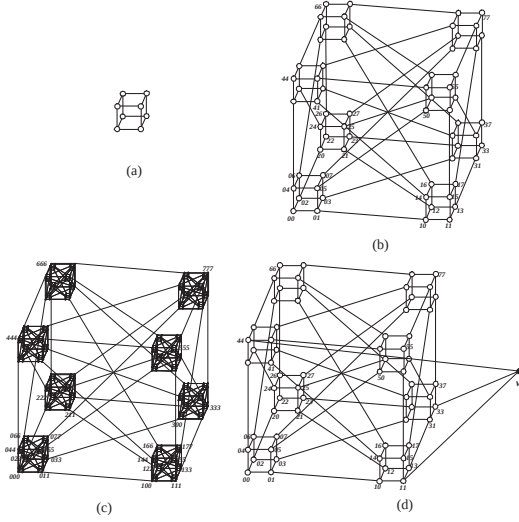


Figure 1: The graphs (a) $FCCN_1$, (b) $FCCN_2$, (c) $FCCN_3$, and (d) $FCCN_2^{1,3,4}$

the set $\{p^n \mid p \in \{r, s, t\}\}$ to an extra vertex w . For example, $FCCN_2^{1,3,4}$ is illustrated in Figure 1(d). By the complete property, we only discuss $FCCN_n^{0,1,2}$.

The following theorem is proved in [5].

Theorem 1 [5] *$FCCN_n$ is hamiltonian, hamiltonian connected for $n \geq 2$.*

3 Main result

Theorem 2 *The n -dimensional fully connected cubic network $FCCN_n$ is 3*-connected for $n \geq 2$.*

Proof. For the ease of proving, we will show both $FCCN_n$ and $FCCN_n^{0,1,2}$ are 3*-connected for $n \geq 2$ in the following.

We prove the statement by induction on n . Since $FCCN_2$ has 64 vertices, $FCCN_2^{0,1,2}$ has 65 vertices, we can check that $FCCN_2$, $FCCN_2^{0,1,2}$ are 3*-connected by brute force.

We assume the statement holds for $FCCN_l$ and $FCCN_l^{0,1,2}$ with $2 \leq l \leq n$.

First, we prove that $FCCN_{n+1}$ is 3*-connected. Let u and v be any two distinct vertices in $FCCN_{n+1}$. We need to find a 3*-container $C^*(u, v)$ of $FCCN_{n+1}$ joining u to v .

Suppose that $u \in bFCCN_n$ and $v \in eFCCN_n$ with $b \neq e$. Without loss of generality, we assume that $b = 0$ and $e = 7$. By induction, there exists a 3*-container $C_0^*(u, w)$ joining u to w in $0FCCN_n^{1,2,7}$ and there exists a 3*-container $C_7^*(v, w)$ joining v to w in $7FCCN_n^{0,1,6}$. Let $C_0^*(u, w) = \{\langle u, P_{01}, 07^n, w \rangle, \langle u, P_{02}, 01^n, w \rangle, \langle u, P_{03}, 02^n, w \rangle\}$ and $C_7^*(v, w) = \{\langle v, P_{71}, 70^n, w \rangle, \langle v, P_{72}, 71^n, w \rangle, \langle v, P_{73}, 76^n, w \rangle\}$. Let $r_i = i(i-1)^n$ for $3 \leq i \leq 6$, $s_j = j(j+1)^n$ for $2 \leq j \leq 6$, and $r_2 = 20^n$. By Theorem 1, there exists a hamiltonian path P_1 joining 10^n to 17^n in $1FCCN_n$, and there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $2 \leq i \leq 6$. Let $P_1^* = \langle u, P_{01}, 07^n, 70^n, P_{71}, v \rangle$, $P_2^* = \langle u, P_{02}, 01^n, 10^n, P_1, 17^n, 71^n, P_{72}, v \rangle$, and $P_3^* = \langle u, P_{03}, 02^n, 20^n, P_2, P_3, \dots, P_6, 67^n, 76^n, P_{73}, v \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(a) for an illustration.

Suppose that $\{u, v\} \subset bFCCN_n$. Without loss of generality, we assume that $b = 0$. By induction, there exists a 3*-container $C_0^*(u, w) = \{P_{01}, P_{02}, P_{03}\}$ joining u to w in $0FCCN_n^{1,2,7}$. We assume that P_{03} be the path contain vertex w . Obviously, P_{03} can be written as $\langle u, P'_{03}, 0x^n, w, 0y^n, P''_{03}, v \rangle$ where $|\{x, y\} \cap \{1, 2, 7\}| = 2$. Without loss of generality, we assume that $x = 1$ and $y = 7$. Let $r_i = i(i-1)^n$ for $1 \leq i \leq 7$, $s_j = j(j+1)^n$ for $1 \leq j \leq 6$ and $s_7 = 70^n$. By Theorem 1, there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $1 \leq i \leq 7$. Let $P_3^* = \langle u, P'_{03}, 01^n, P_1, P_2, \dots, P_7, 07^n, P''_{03}, v \rangle$. Obviously, $C^*(u, v) = \{P_{01}, P_{02}, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(b) for an illustration.

Second, we prove that $FCCN_{n+1}^{0,1,2}$ is 3*-connected.

Let u and v be any two distinct vertices in $FCCN_{n+1}^{0,1,2}$. We need to find a 3*-container $C^*(u, v)$ of $FCCN_{n+1}^{0,1,2}$ joining u to v .

Suppose that $u \in bFCCN_n$ and v is the extra vertex w . If $b \in \{0, 1, 2\}$. Without loss of generality, we assume that $b = 0$. By induction, there exists a 3*-container $C_0^*(u, w) = \{\langle u, P_{01}, 00^n, w \rangle, \langle u, P_{02}, 01^n, w \rangle, \langle u, P_{03}, 07^n, w \rangle\}$ joining u to w in $0FCCN_n^{0,1,7}$. Let $r_i = i(i+1)^n$ for $2 \leq i \leq 6$, $s_i = j(j-1)^n$ for $3 \leq j \leq 7$, $s_2 = 22^n$, and $r_7 = 70^n$. By Theorem 1, there exists a hamiltonian path P_1 joining 10^n to 11^n

in $1FCCN_n$ and there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $2 \leq i \leq 7$. Let $P_1^* = \langle u, P_{01}, 00^n, w \rangle$, $P_2^* = \langle u, P_{02}, 01^n, P_1, w \rangle$, and $P_3^* = \langle u, P_{03}, 07^n, P_7, P_6, \dots, P_2, 22^n, w \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(c) for an illustration. If $b \notin \{0, 1, 2\}$, by the complete property, it is easy to check that $b \in N(0)$. Without loss of generality, we assume that $b = 3$. By induction, there exists a 3*-container $C_3^*(u, w) = \{\langle u, P_{31}, 30^n, w \rangle, \langle u, P_{32}, 31^n, w \rangle, \langle u, P_{33}, 37^n, w \rangle\}$ joining u to w in $3FCCN_n^{0,1,7}$. Let $r_i = i(i+1)^n$ for $4 \leq i \leq 6$, $s_i = j(j-1)^n$ for $5 \leq j \leq 7$, $s_4 = 42^n$, and $r_7 = 73^n$. By Theorem 1, there exists a hamiltonian path P_0 joining 00^n to 03^n in $0FCCN_n$, there exists a hamiltonian path P_1 joining 13^n to 11^n in $1FCCN_n$, there exists a hamiltonian path P_2 joining 24^n to 22^n in $2FCCN_n$, and there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $4 \leq i \leq 7$. Let $P_1^* = \langle u, P_{31}, 03^n, P_0, 00^n, v \rangle$, $P_2^* = \langle u, P_{32}, 13^n, P_1, 11^n, v \rangle$, and $P_3^* = \langle u, P_{33}, 37^n, P_7, P_6, \dots, P_4, 42^n, 24^n, P_2, 22^n, v \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(d) for an illustration.

Suppose that $u \in bFCCN_n$ and $v \in eFCCN_n$ with $b \neq e$. If $|\{b, e\} \cap \{0, 1, 2\}| = 2$, without loss of generality, we assume that $b = 0$ and $e = 1$. By induction, there exists a 3*-container $C_0^*(u, w) = \{\langle u, P_{01}, 01^n, w \rangle, \langle u, P_{02}, 00^n, w \rangle, \langle u, P_{03}, 07^n, w \rangle\}$ joining u to w in $0FCCN_n^{0,1,7}$, there exists a 3*-container $C_1^*(v, w) = \{\langle v, P_{10}, 10^n, w \rangle, \langle v, P_{12}, 11^n, w \rangle, \langle v, P_{13}, 12^n, w \rangle\}$ joining v to w in $1FCCN_n^{0,1,2}$. Let $r_i = i(i+1)^n$ for $2 \leq i \leq 6$, $s_i = j(j-1)^n$ for $2 \leq j \leq 7$, and $r_7 = 70^n$. By Theorem 1, there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $2 \leq i \leq 7$. Let $P_1^* = \langle u, P_{01}, 01^n, 10^n, P_{11}, v \rangle$, $P_2^* = \langle u, P_{02}, 00^n, w, 11^n, P_{12}, v \rangle$, and $P_3^* = \langle u, P_{03}, 07^n, 70^n, P_7, P_6, \dots, P_2, 21^n, 12^n, P_{13}, v \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(e) for an illustration.

If $|\{b, e\} \cap \{0, 1, 2\}| = 1$, without loss of generality, we assume that $b = 0$ and $e = 3$. By induction, there exists a 3*-container $C_0^*(u, w) = \{\langle u, P_{01}, 00^n, w \rangle, \langle u, P_{02}, 02^n, w \rangle, \langle u, P_{03}, 07^n, w \rangle\}$ joining u to w in $0FCCN_n^{0,2,7}$, there exists a 3*-container $C_3^*(v, w) = \{\langle v, P_{31}, 31^n, w \rangle, \langle v, P_{32}, 32^n, w \rangle, \langle v, P_{33}, 34^n, w \rangle\}$ joining v to w in

$3FCCN_n^{1,2,4}$. Let $r_i = i(i+1)^n$ for $4 \leq i \leq 6$, $s_i = j(j-1)^n$ for $4 \leq j \leq 7$, and $r_7 = 70^n$. By Theorem 1, there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $4 \leq i \leq 7$, there exists a hamiltonian path P_1 joining 11^n to 13^n in $1FCCN_n$, there exists a hamiltonian path P_2 joining 20^n to 23^n in $2FCCN_n$. Let $P_1^* = \langle u, P_{01}, 00^n, w, 11^n, P_1, 13^n, 31^n, P_{31}, v \rangle$, $P_2^* = \langle u, P_{02}, 02^n, 20^n, P_2, 23^n, 32^n, P_{32}, v \rangle$, and $P_3^* = \langle u, P_{03}, 07^n, 70^n, P_7, P_6, \dots, P_4, 43^n, 34^n, P_{33}, v \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(f) for an illustration.

If $|\{b, e\} \cap \{0, 1, 2\}| = 0$, without loss of generality, we assume that $b = 3$ and $e = 4$. By induction, there exists a 3*-container $C_3^*(u, w) = \{\langle u, P_{31}, 30^n, w \rangle, \langle u, P_{32}, 32^n, w \rangle, \langle u, P_{33}, 37^n, w \rangle\}$ joining u to w in $3FCCN_n^{0,2,7}$, there exists a 3*-container $C_4^*(v, w) = \{\langle v, P_{41}, 41^n, w \rangle, \langle v, P_{42}, 42^n, w \rangle, \langle v, P_{43}, 45^n, w \rangle\}$ joining v to w in $4FCCN_n^{1,2,5}$. By Theorem 1, there exists a hamiltonian path P_0 joining 03^n to 00^n in $0FCCN_n$, there exists a hamiltonian path P_1 joining 11^n to 14^n in $1FCCN_n$, there exists a hamiltonian path P_2 joining 23^n to 24^n in $2FCCN_n$, there exists a hamiltonian path P_5 joining 54^n to 56^n in $5FCCN_n$, there exists a hamiltonian path P_6 joining 65^n to 67^n in $6FCCN_n$, there exists a hamiltonian path P_7 joining 76^n to 73^n in $3FCCN_n$. Let $P_1^* = \langle u, P_{31}, 30^n, 03^n, P_0, w, 11^n, P_1, 14^n, 41^n, P_{41}, v \rangle$, $P_2^* = \langle u, P_{32}, 32^n, 23^n, P_2, 24^n, 42^n, P_{42}, v \rangle$, and $P_3^* = \langle u, P_{33}, 37^n, 73^n, P_7, P_6, P_5, 54^n, 45^n, P_{43}, v \rangle$. Obviously, $C^*(u, v) = \{P_1^*, P_2^*, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(g) for an illustration.

Suppose that $\{u, v\} \subset bFCCN_n$. If $b \in \{0, 1, 2\}$, without loss of generality, we assume that $b = 0$. By induction, there exists a 3*-container $C_0^*(u, w) = \{P_{01}, P_{02}, P_{03}\}$ joining u to w in $0FCCN_n^{1,2,7}$. We assume that P_{03} be the path contain vertex w . Obviously, P_{03} can be written as $\langle u, P'_{03}, 0x^n, w, 0y^n, P''_{03}, v \rangle$ where $|\{x, y\} \cap \{1, 2, 7\}| = 2$. Without loss of generality, we assume that $x = 1$ and $y = 7$. Let $r_i = i(i-1)^n$ for $3 \leq i \leq 7$, $s_j = j(j+1)^n$ for $3 \leq j \leq 6$ and $s_7 = 70^n$. By Theorem 1, there exists a hamiltonian path P_1 joining 10^n to 11^n in $1FCCN_n$, there exists a hamiltonian path P_2 joining 22^n to 23^n in $2FCCN_n$, there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $3 \leq i \leq 7$. Let $P_3^* =$

$\langle u, P'_{03}, 01^n, 10^n, P_1, 11^n, w, 22^n, P_2, 23^n, 32^n, P_3, \dots, P_7, 07^n, P''_{03}, v \rangle$. Obviously, $C^*(u, v) = \{P_{01}, P_{02}, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(h) for an illustration.

If $b \notin \{0, 1, 2\}$, without loss of generality, we assume that $b = 7$. By induction, there exists a 3*-container $C_7^*(u, w) = \{P_{71}, P_{72}, P_{73}\}$ joining u to w in $7FCCN_n^{0,1,2}$. We assume that P_{03} be the path contain vertex w . Obviously, P_{03} can be written as $\langle u, P'_{73}, 7x^n, w, 7y^n, P''_{73}, v \rangle$ where $|\{x, y\} \cap \{0, 1, 2\}| = 2$. Without loss of generality, we assume that $x = 0$ and $y = 2$. Let $r_i = i(i-1)^n$ for $4 \leq i \leq 6$, $s_j = j(j+1)^n$ for $3 \leq j \leq 5$, $r_3 = 31^n$, and $s_6 = 62^n$. By Theorem 1, there exists a hamiltonian path P_0 joining 07^n to 00^n in $0FCCN_n$, there exists a hamiltonian path P_1 joining 11^n to 13^n in $1FCCN_n$, there exists a hamiltonian path P_2 joining 26^n to 27^n in $2FCCN_n$, there exists a hamiltonian path P_i joining r_i to s_i in $iFCCN_n$ for $3 \leq i \leq 6$. Let $P_3^* = \langle u, P'_{73}, 70^n, 07^n, P_0, 00^n, w, 11^n, P_1, 13^n, 31^n, P_3, \dots, P_6, P_2, 27^n, 72^n, P''_{73}, v \rangle$. Obviously, $C^*(u, v) = \{P_{71}, P_{72}, P_3^*\}$ forms 3*-container of $FCCN_{n+1}$ joining u to v . See Figure 2(i) for an illustration.

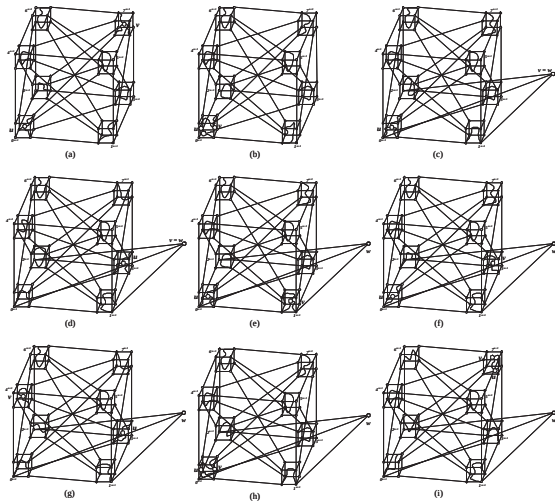


Figure 2: Illustration for Theorem 2

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